

Convolutional Neural Networks:

$$H_{in} \times W_{in} \times D_{in} \rightarrow 5 \times 5 \times 3$$

$$\text{Number of Filters : } K : 2$$

$$\text{Spatial Extent : } F : 3$$

$$\text{Stride : } S : 2$$

$$\text{Padding : } P : 1$$

$$H_{in} \times W_{in} \times D_{in} \xrightarrow[\text{(K, F, P, S)}]{\text{Conv}} H_{out} \times W_{out} \times D_{out}$$

$$H_{out} = \left(\frac{H_{in} - F + 2P}{S} \right) + 1 \quad \left(\frac{5 - 3 + 2}{2} \right) + 1 = 3$$

$$W_{out} = \left(\frac{W_{in} - F + 2P}{S} \right) + 1 \quad \left(\frac{5 - 3 + 2}{2} \right) = 3$$

$$D_{out} = K = 2$$

$$H_{out} \times W_{out} \times D_{out} : 3 \times 3 \times 2$$



